

# Project Requirements Document (PRD)

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**Project Title:** Skill-Exchange (Barter-Based Service App)

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## 1. Introduction

Skill-Exchange is an Android application designed to enable rural technicians to exchange services using a barter system. Instead of money, users trade skills based on time (skill points), building a collaborative and self-reliant ecosystem.

## 2. Problem Statement

Rural technicians (plumbers, electricians, carpenters, etc.) often need help from other skilled workers but lack the financial means to pay. Additionally, there is no structured platform to discover available skills within the community. This leads to underutilization of skills and limited collaboration.

## 3. Scope of the App

### In Scope

- User registration and authentication
- Skill profile creation and management
- Posting service requests (Need Posts)
- Responding with swap offers
- Skill-based filtering of posts
- Skill point system (1 hour = 1 point)
- Trust score system based on completed exchanges

- Chat interface for negotiation
- Real-time data updates

### **Out of Scope**

- Monetary transactions or payment gateways
- Advanced AI-based recommendations (initial version)
- Multi-language support (future enhancement)
- Integration with external service providers

## **4. Features**

### **Must-Have Features (Core Requirements)**

- User login/signup
- Create and edit skill profile
- Post “Need” requests with required skill
- Filter posts by skill
- Send and receive swap offers
- Skill point system (time-based exchange)
- Trust score update after mutual confirmation
- Basic chat functionality between users

### **Good-to-Have Features (Enhancements)**

- AI-based skill matching suggestions
- Voice input for low-literacy users
- Multi-language support (regional languages)
- Ratings and reviews system
- Notifications for new offers/messages
- Offline support (limited functionality)

## **5. Functional Requirements**

### **5.1 User Management**

- Users must be able to register and log in
- Users can create, view, and update their profiles

### **5.2 Skill Profile**

- Users must specify their primary skill(s)
- Skills should be searchable and filterable

### **5.3 Need Post**

- Users can create posts describing their requirements
- Each post must include:
  - Description
  - Required skill
  - Estimated time (hours)

### **5.4 Swap Offer**

- Users can respond to posts with service exchange offers
- Offers include skill offered and time commitment

### **5.5 Skill Point System**

- 1 hour of work = 1 point
- Points are transferred after successful exchange

### **5.6 Trust Score System**

- Trust score increases only when both users confirm completion
- Prevents misuse and builds reliability

### **5.7 Chat System**

- Users can communicate within the app to negotiate terms

### **5.8 Filtering System**

- Users must be able to filter posts based on required skill

## 6. Non-Functional Requirements

- **Usability:** Simple and intuitive UI suitable for rural users
- **Performance:** Real-time updates using Firebase Firestore
- **Reliability:** Accurate tracking of points and trust score
- **Scalability:** Able to handle increasing users and data
- **Security:** Secure authentication and basic data protection
- **Availability:** System should be accessible with minimal downtime

## 7. Technology Stack (Overview)

- **Frontend:** Android (Jetpack Compose)
- **Backend:** Firebase
- **Database:** Firebase Firestore
- **Authentication:** Firebase Authentication

## 8. Conclusion

The Skill-Exchange app provides a practical solution to enable skill-based collaboration without financial dependency. By combining a barter system with digital accessibility, it strengthens community bonds and promotes self-sufficiency among rural technicians.